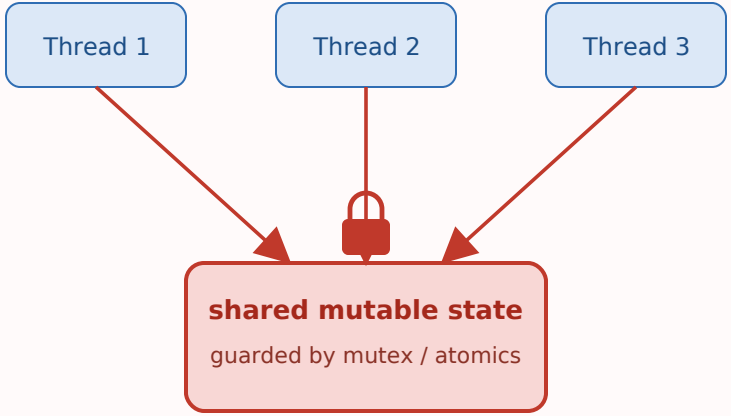
# Two concurrency models for control-plane state

## Shared-memory concurrency

threads mutate common state behind locks



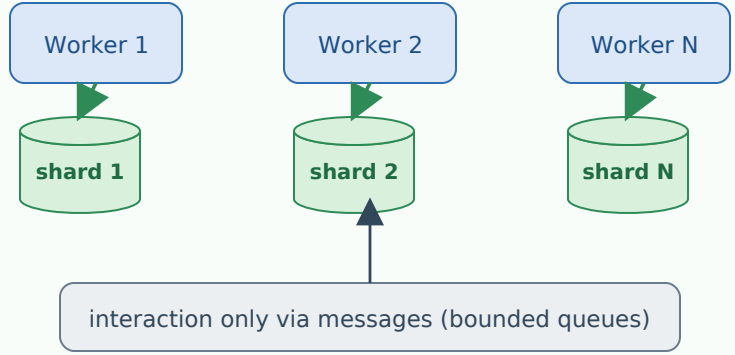

**Hazards:**

data races · deadlock · lock contention
priority inversion · hard-to-reason worst case

*worst-case timing is hard to bound*

## Shared-nothing (message-passing)

state partitioned, each piece owned by one thread




**Benefits:**

no shared mutable state · no locks on data
simpler reasoning · scales · bounded worst case

**chosen for BOTH implementations (maps to per-UE state)**